University of Zurich[UZH] | ETH zürich

Institute of Informatics – Institute of Neuroinformatics

ROBOTICS & PERCEPTION GROUP

Vision-based Navigation:

from State Estimation to Motion Planning

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02.02.2020 @深蓝学院

About Me

Zichao Zhang（张子潮)

Education

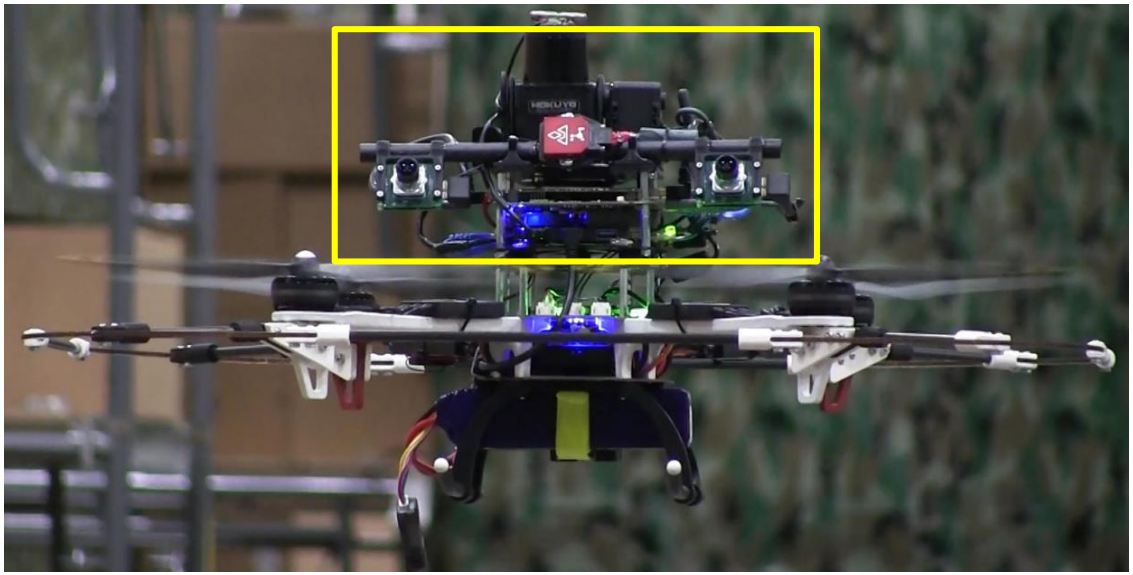
- Bachelor: Beihang University
- Master: Department of Informatics, University of Zurich
- Currently a PhD student at Robotics and Perception Group, University of Zurich and ETH Zurich

Research interest

- Visual-based navigation
- Sensor fusion
- Active vision

Personal page GitHub Scholar

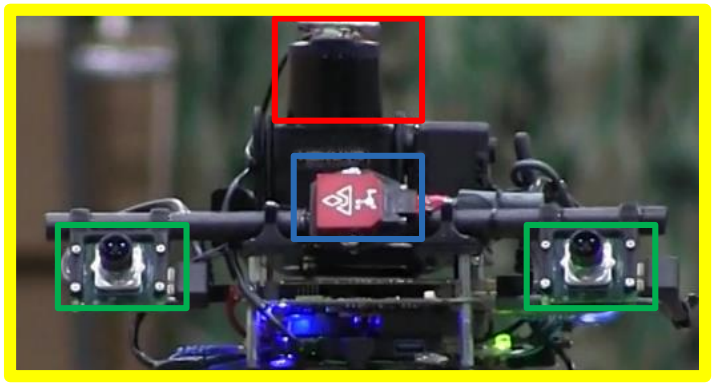
Vision Algorithms in Autonomous Agents



Autonomous flight in GPS-denied environment requires complicated sensor systems.

DARPA Fast Light Weight Autonomy Program

Mohta et al, Fast, autonomous flight in GPS-denied and cluttered environments, *JFR 2016.*

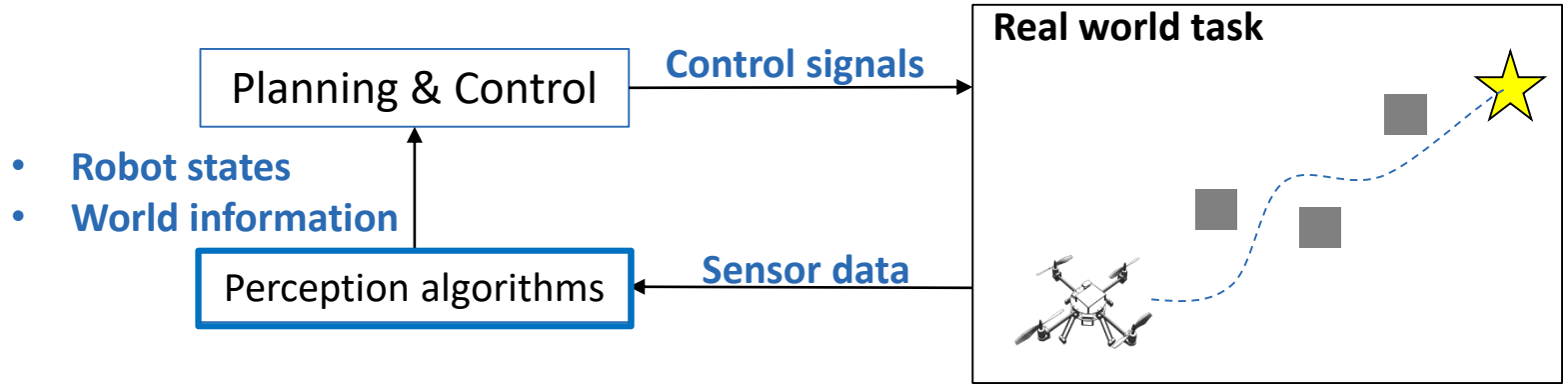


Stereo cameras

Inertial measurement unit

Lidar

Vision Algorithms in Autonomous Agents



Vision algorithms are responsible for many tasks:

- Depth sensing for obstacle avoidance
- Dense (or higher-level mapping) for general planning tasks
- **Real-time vision-based state estimation**

  Estimate the robot's ego-motion using image streams from the cameras (possibly combined with other sensors).

  There are many variants and names:
  - Visual(-inertial) odometry
  - Visual(-inertial) SLAM
  - ……

Outline

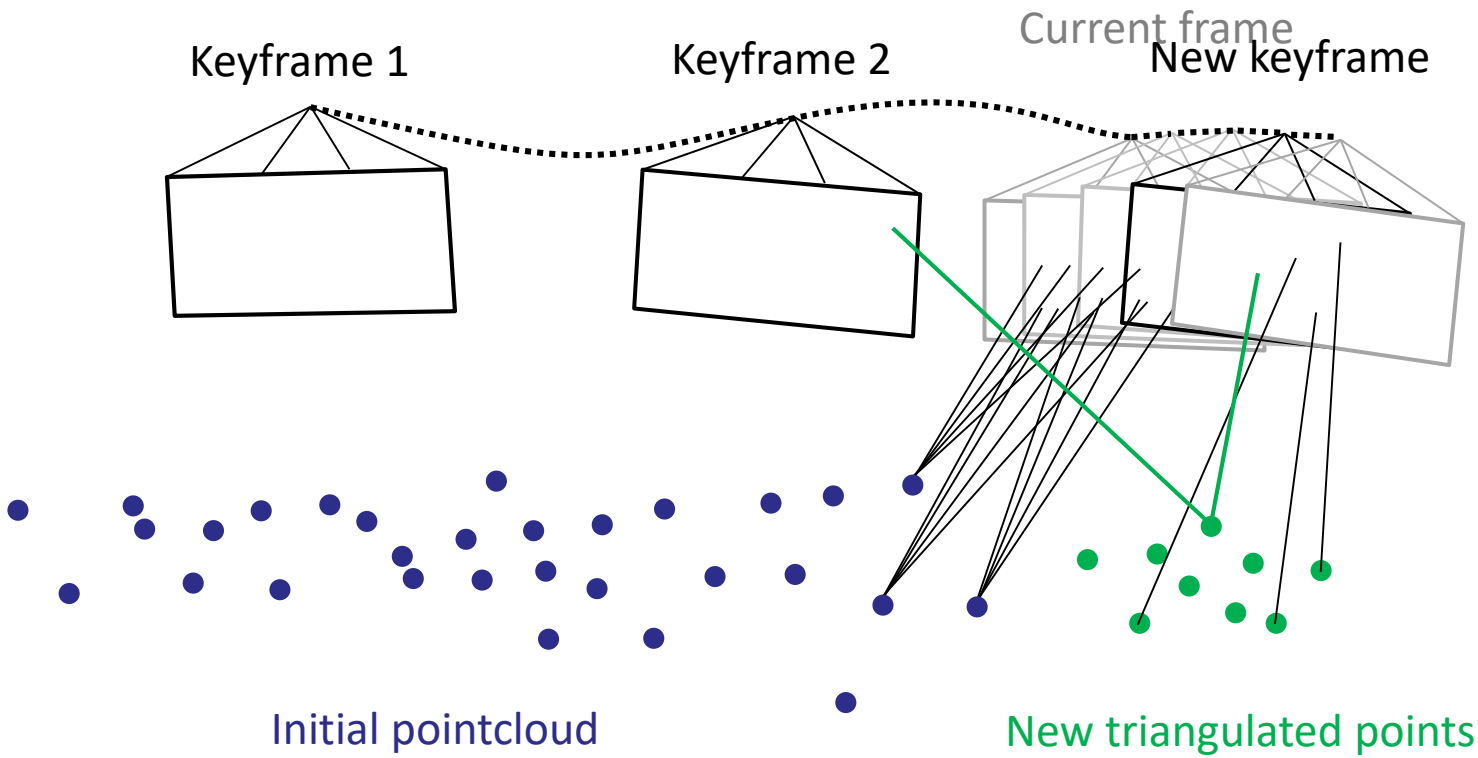
- Vision-based state estimation
  - A short introduction to visual SLAM
  - Vision-based state estimation for mobile robots
- Perception-aware motion planning
  - Fisher information and its application
  - Receding horizon planning and map representation
- Outlook

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Vision-based state estimation

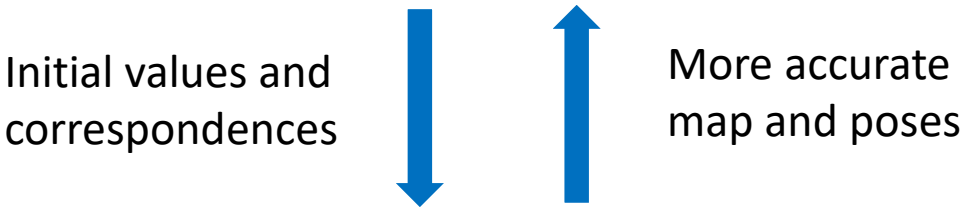
- Frontend (visual odometry): **alternate** pose estimation and triangulation



Vision-based state estimation

- Frontend (visual odometry): **alternating estimation**
  - (Initial) pose estimation for incoming images (also known as tracking)
  - Creation and management of the map (e.g., 3D points)
    - E.g., keyframe selection --> extending the current map when it is not enough for tracking

  There are different methods that can be used to accomplish the above tasks (direct, feature-based; dense, sparse, etc.)



- Backend: **jointly** optimize camera poses and points
  - Bundle adjustment (with sparsification or marginalization)
  - Incorporate loop closures → Simultaneously Localization and Mapping (SLAM)

- Incorporate other sensors (e.g., IMU) in both the frontend and the backend.

Common building blocks in visual SLAM

A visual SLAM pipeline makes use of many computer vision algorithms.

- Data association
  - Feature detection, extraction and matching
  - Template matching (e.g., Lucas-Kanade tracking)
- Motion estimation
  - Relative pose estimation
  - Absolute pose estimation (e.g., perspective-n-points problem)
  - Place recognition and visual localization
- Mapping
  - Triangulation (2 views or multiple views)
  - Bundle adjustment and variants

→ **What are the differences in the context of SLAM (compared with pure vision problems)?**

The benefits and limitations

- We usually have (very) strong priors

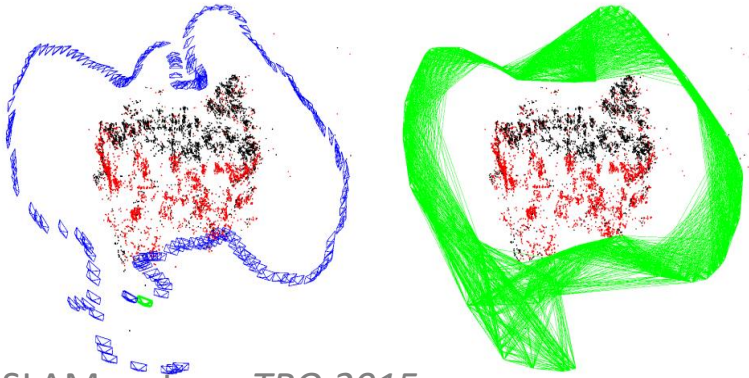
  - Frame-to-frame motion is often small (easier to estimate).
  - Co-visibility constraints provide good candidates for data association.
  - Visual frontends (combined with other sensors) provide good initial values for the backend (nonlinear least-squares) optimization.

Example: Data association in SLAM

- Parallel Tracking and Mapping (PTAM)
  - Use motion prior for the current frame (decay velocity)
  - Find a small subset of matches first, estimate a coarse pose, and search for more matches from the coarse pose.

Klein et al, Parallel tracking and mapping for small AR workspaces, *ISMAR 2007.*

- ORB-SLAM
  - Constant velocity motion prior
  - Find initial matches from last frame
  - Find more matches from co-visible keyframes



Mur-Artal et al, ORB-SLAM: a versatile and accurate monocular SLAM system, *TRO 2015.*

Similar tricks can be seen in many SLAM pipelines...

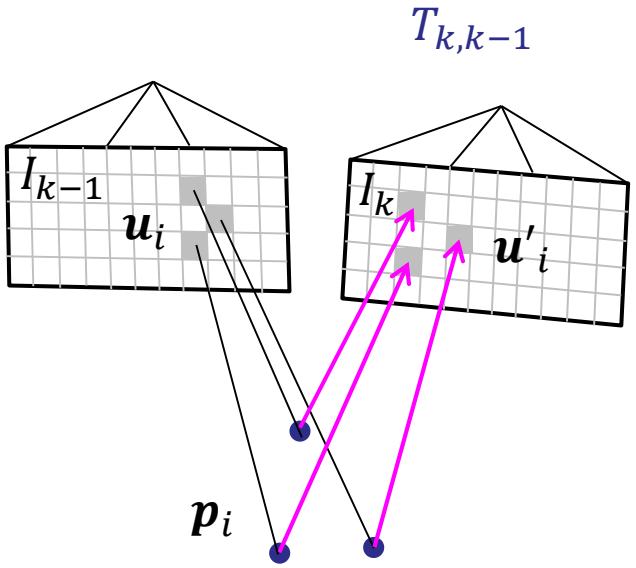
Without motion prior, the tracking (motion estimation) is essentially a visual localization problem, which is not easy to solve in real-time.

Benchmark Long-term Visual Localization, https://www.visuallocalization.net/

Example: Utilizing small intra-frame motion

- (model-based) Direct image alignment

$$T_{k,k-1} = \arg\min_{T} \sum_i \| I_k(\boldsymbol{u}'_i) - I_{k-1}(\boldsymbol{u}_i) \|^2$$



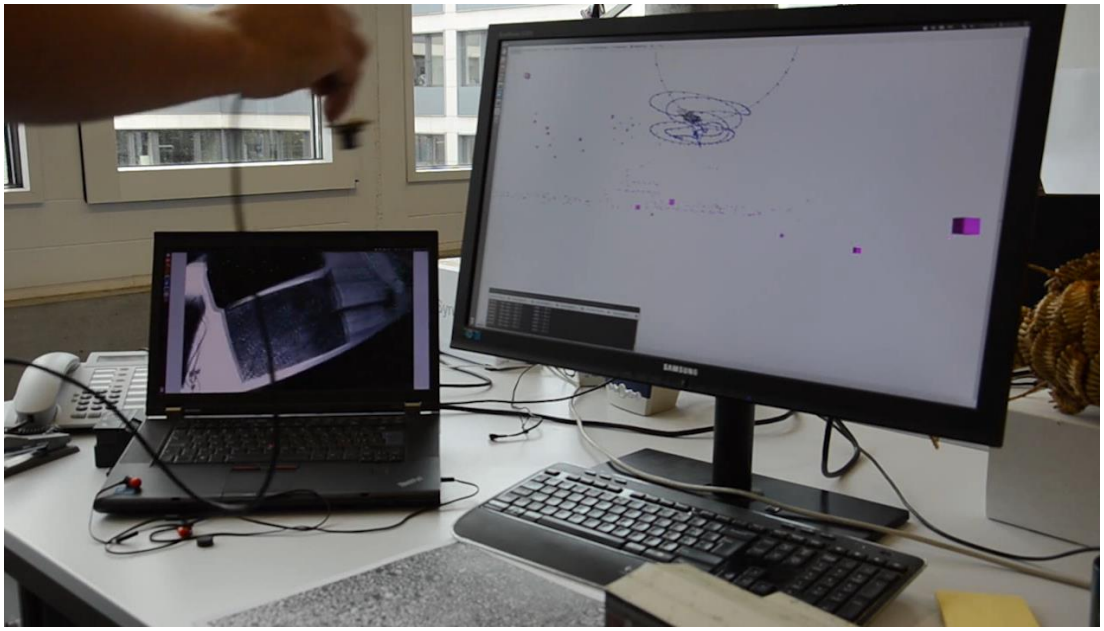
- Converges fast when the relative motion is small.
- Not robust to large frame motion.

Works well on a high frame-rate camera (align every frame w.r.t. the previous one).

Example: Utilizing small intra-frame motion

- (model-based) Direct image alignment
  - Converges fast when the relative motion is small.
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Works well on a high frame-rate camera (align every frame w.r.t. the previous one).



(estimate fast motion with a single camera, no IMU)

A key component in SVO:

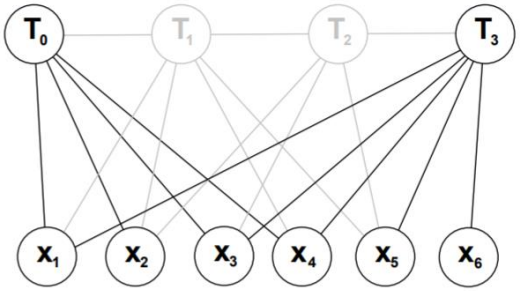
Provides an efficient and robust way to get motion prior for later data association and motion estimation.

Forster et al, SVO: Semidirect visual odometry for monocular and multicamera systems, *TRO 2016.*

The benefits and limitations

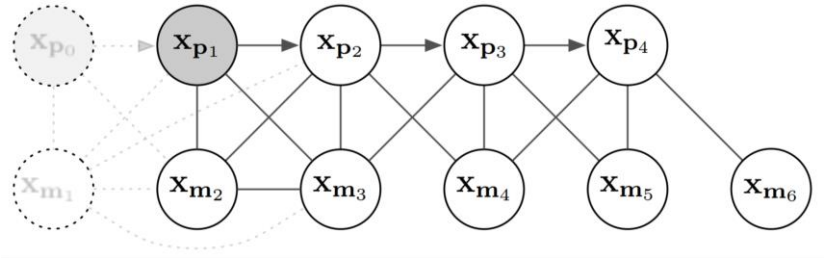
- We usually have (very) strong priors
- Limited computational resource (e.g., process time/frame, memory)
  - How to reduce the computational cost without hurting (too much) the accuracy?

- Selectively discard or marginalize information



Take a sparse subset of keyframes

Strasdat et al, Visual SLAM: Why Filter? 2012



Marginalize/discard oldest frames

Sibley et al, Sliding Window Filter with Application to Planetary Landing, JFR 2010

- Parallel threads for different tasks

Fast	Frame-rate motion estimation
Slow	Bundle adjustment in a local window
Slower	Loop closure and global bundle adjustment

What are special for mobile robots?

- More restricted computational resource (also power) ☹
- Also many more interesting dimensions to explore ☺
  - Access to various sensors
    - Combining different / multiple sensors (visual-inertial, multi-camera)



Skydio R2 (6 fisheye cameras)

Picture from: The Verge

- More cameras brings more information
  If handled properly, they should always bring better performance.
- However, it complicates the system design (**more on this later**)

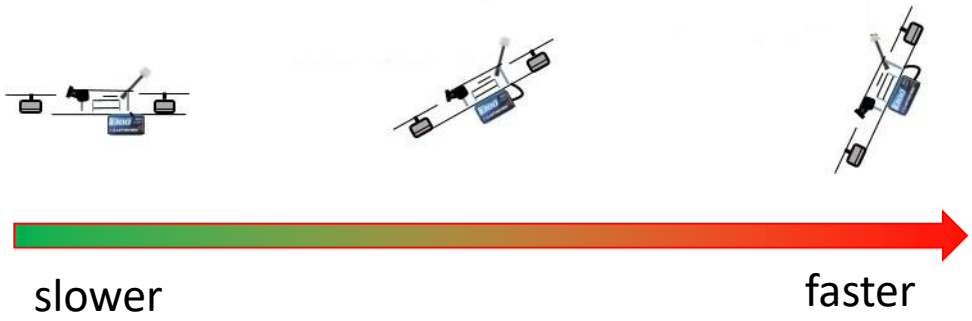
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    - Sensor placement matters



Tilted camera on an FPV drone

Picture from: www.getfpv.com



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- Also many more interesting dimensions to explore ☺
  - Access to various sensors
    - Combining different / multiple sensors (visual-inertial, multi-camera)
    - Sensor placement matters
    - (automatic) sensor configuration
      - Exposure/gain
        Zhang et al, Active Exposure Control for Robust Visual Odometry in HDR Environments, *ICRA2017.*
      - Camera field-of-view
        Zhang et al, Benefit of Large Field-of-View Cameras for Visual Odometry, *ICRA2016.*

What are special for mobile robots?

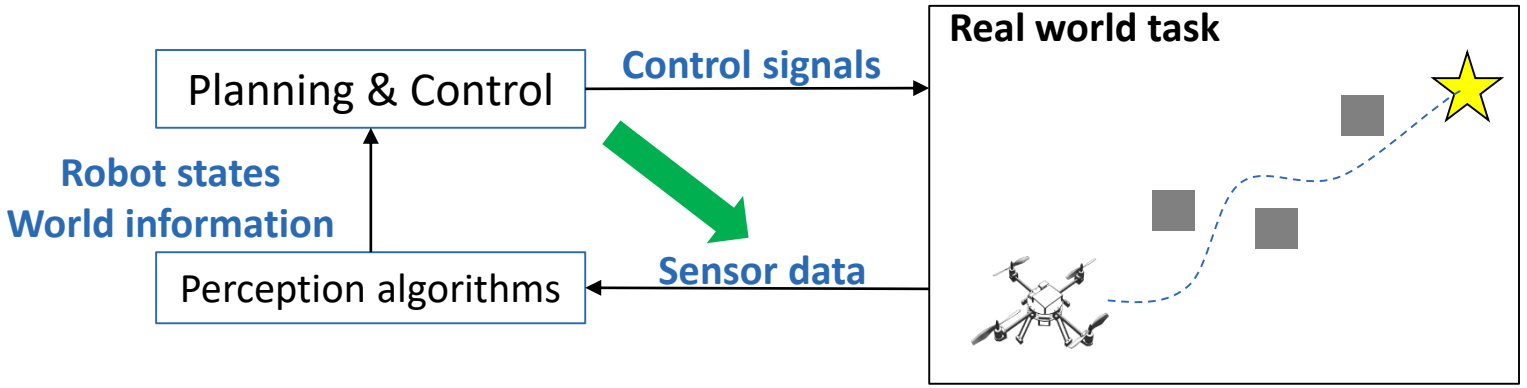
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- Also many more interesting dimensions to explore ☺
  - Access to various sensors
    - Combining different / multiple sensors (visual-inertial, multi-camera)
    - Sensor placement matters
    - (automatic) sensor configuration
  - **The coupling between perception and robot motion**
    - How does the motion of the robot influence the perception algorithms? How should we make use of it?

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Active SLAM/Localization Problem

- The motion of the robot influences the sensor data



Active Perception: Need to consider the impact of the robot motion on perception algorithms at the planning stage to better accomplish a task.

- For SLAM/localization: how to quantify the quality of visual localization/vision-based motion estimation?
  - Heuristics: number of features, richness in texture…
  - Information-theoretic metrics: Fisher information

SLAM as a Nonlinear Least Squares Problem

- Independent measurements → Maximum a Posteriori

State (pose, velocity, etc)

$$P(X, L|Z) \propto P(x_0)\left(\prod_{i=1}^{M} P(z_i|x_{i_k}, l_{i_j})\right)\left(\prod_{k=2}^{N} P(x_k|x_{k-1})\right)$$

Landmarks

Measurements (visual, inertial)

Visual measurements

Kinematic constraints (motion model, IMU)

- Gaussian noise → Nonlinear Least Squares (iterative optimization)

Without the prior, applying the property of Gaussian distribution:

$$\{X^*, L^*\} = \underset{\{X,L\}}{\operatorname{argmax}} P(X, L|Z)$$

$$= \underset{\{X,L\}}{\operatorname{argmin}} \left\{ \sum_{k=1}^{N} \|f(x_{k-1}) - x_k\|_{\Lambda_k}{}^2 + \sum_{i=1}^{M} \left\| h(x_{i_k}, l_{i_j}) - z_i \right\|_{\Sigma_i}{}^2 \right\}$$

NLLS, Fisher Information and Covariance

➢ Solving NLLS with Gauss-Newton method

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \sum_{i=1}^{M} \|f_i(\boldsymbol{\theta}) - \mathbf{z}_i\|_{\Sigma_i}^2$$

Solve it iteratively (rotation: local coordinates)

$$\boldsymbol{\varepsilon}^* = \underset{\varepsilon}{\operatorname{argmin}} \sum_{i=1}^{M} \|f_i(\boldsymbol{\theta}^s + \boldsymbol{\varepsilon}) - \mathbf{z}_i\|_{\Sigma_i}^2 \qquad \boldsymbol{\theta}^{s+1} \leftarrow \boldsymbol{\theta}^s + \boldsymbol{\varepsilon}$$

Applying first-order approximation:

$$\boldsymbol{\varepsilon}^* = -(\mathbf{J}^T \Sigma^{-1} \mathbf{J})^{-1} \mathbf{J}^T \mathbf{r}(\boldsymbol{\theta})$$

Jacobian of the measurements w.r.t. parameter to be estimated.

Fisher Information Matrix (FIM)

Some notes:

- The definition of FIM is applicable to general estimation problems (not only NLLS)
- In NLLS (with i.i.d. Gaussian assumption), we get it for free in the solving process.

NLLS, Fisher Information and Covariance

➢ Hessian or Fisher Information Matrix (FIM)

$$\boldsymbol{\varepsilon}^* = -(\mathbf{J}^T\Sigma^{-1}\mathbf{J})^{-1}\mathbf{J}^{\mathrm{T}}\mathbf{r}(\boldsymbol{\theta})$$

- Rank/Null space → what are the (un)observable degrees of freedom?
  - Can be proved to be equivalent to observability gramian

Hinson etal, "Observability-based guidance and sensor placement." PhD diss., 2014.

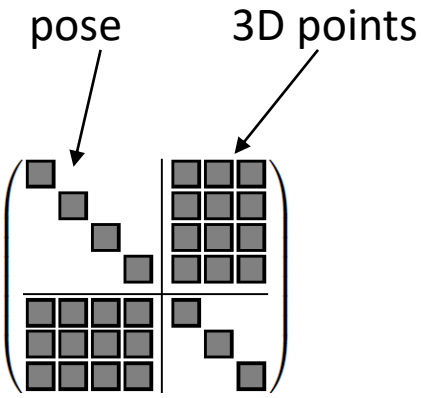
- Inverse → Covariance (lower bound) of the estimate
  - Cramer-Rao lower bound

Hartley etal, Multiple view geometry in computer vision, 2003.

Note: It is only first-order approximation.

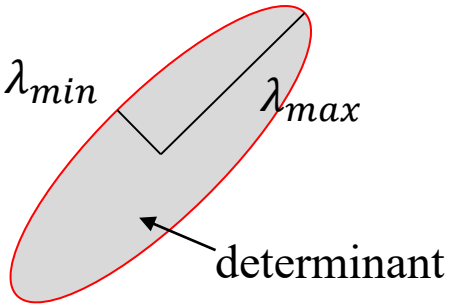
- In bundle adjustment: sparse structure
  - Schur complement
  - (Marginal) dependence of different variables

$$(2\pi)^{-\frac{k}{2}}\det(\boldsymbol{\Sigma})^{-\frac{1}{2}}\,e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$



Information Metric From FIM

- Finally, we need a scalar to say something is good or bad…
- In practice: different scalar metrics can be computed from covariance/FIM to quantify the estimation quality
  - Determinant (differential entropy for Gaussian dist.)
  - Smallest/Largest eigen values
  - Conditional number
  - ……



- Some intuition about the eigen vectors and eigen values

Omitting the measurement noise: $\mathbf{H} = \mathbf{J}^T\mathbf{J}$

J: Jacobian of the measurements w.r.t. parameters to be estimated.

The eigen vectors $\{\boldsymbol{v}_i\}$ of $\mathbf{H}$ is the right singular vector of $\mathbf{J}$:

$$\sigma_i \mathbf{u}_i = \mathbf{J}\boldsymbol{v}_i$$

→ **the mapping from vectors in the parameter space to measurement space**

- If there is some direction $\boldsymbol{v}_{unob}$ in the parameter space that is not in the space spanned by $\{\boldsymbol{v}_i\}$, then $\boldsymbol{v}_{unob}$ is not observable
- The larger $\{\sigma_i\}$, the more observable a direction $\boldsymbol{v}_i$ is (i.e., has more impact on the measurement space)

FIM is a useful tool in robotics/SLAM

- Provides a principled method to quantify the estimation quality
  - Probabilistic interpretation (e.g., KL divergence, different metrics)

- Some applications:

  - Feature selection: select the subset of most informative features

  Carlone et al, Attention and anticipation in fast visual-inertial navigation, TRO 2018

  - Information sparsification/approximation
    - Maintain desired structure (for efficiency) with minimum information loss

  Hsiung et al, Information sparsification in visual-inertial odometry, IROS 2018

  Usenko et al, Visual-inertial mapping with non-linear factor recovery, RAL 2019

  - Optimal sensor placement: where to put sensors on a robot?

  Hinson etal, "Observability-based guidance and sensor placement." PhD diss., 2014.

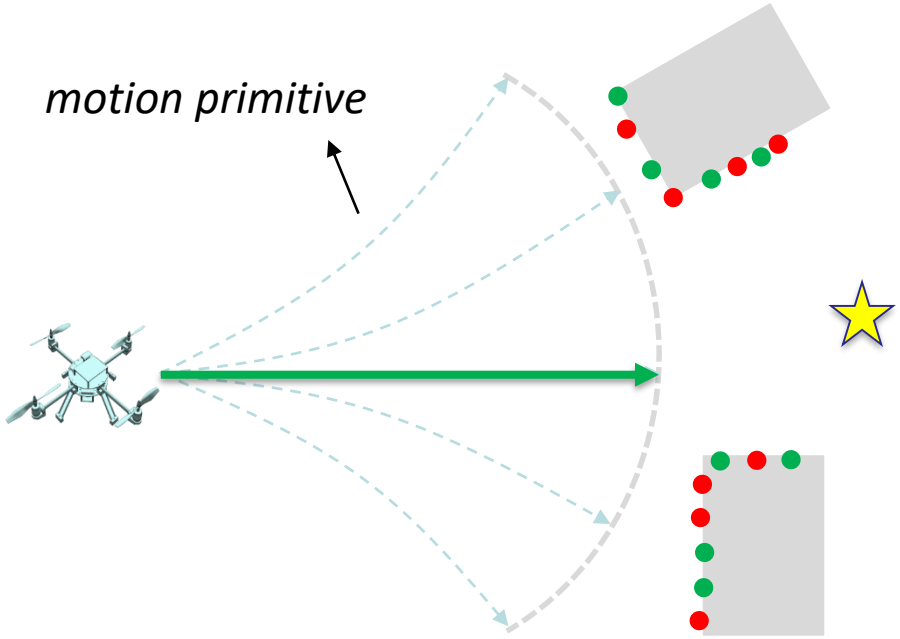
  - Active SLAM and localization

  Core idea: plan the robot motion to maximize some metric from the Fisher information matrix of the SLAM/localization problem.

Perception-aware receding horizon navigation

Zhang et al, Perception-aware Receding Horizon Navigation for MAVs, *ICRA2018.*

- Receding horizon navigation: repeat planning within a limited receding horizon as new information arrives.



1. Use two maps:
   - Dense map: obstacle
   - Active map: SLAM landmarks
2. Generate motion primitives.
3. Evaluate each motion primitive
   - $R_{\mathbf{goal}}$: Progress to the goal
   - $R_{\mathbf{collision}}$: Collision risk
   - $R_{\mathbf{perception}}$: Perception quality

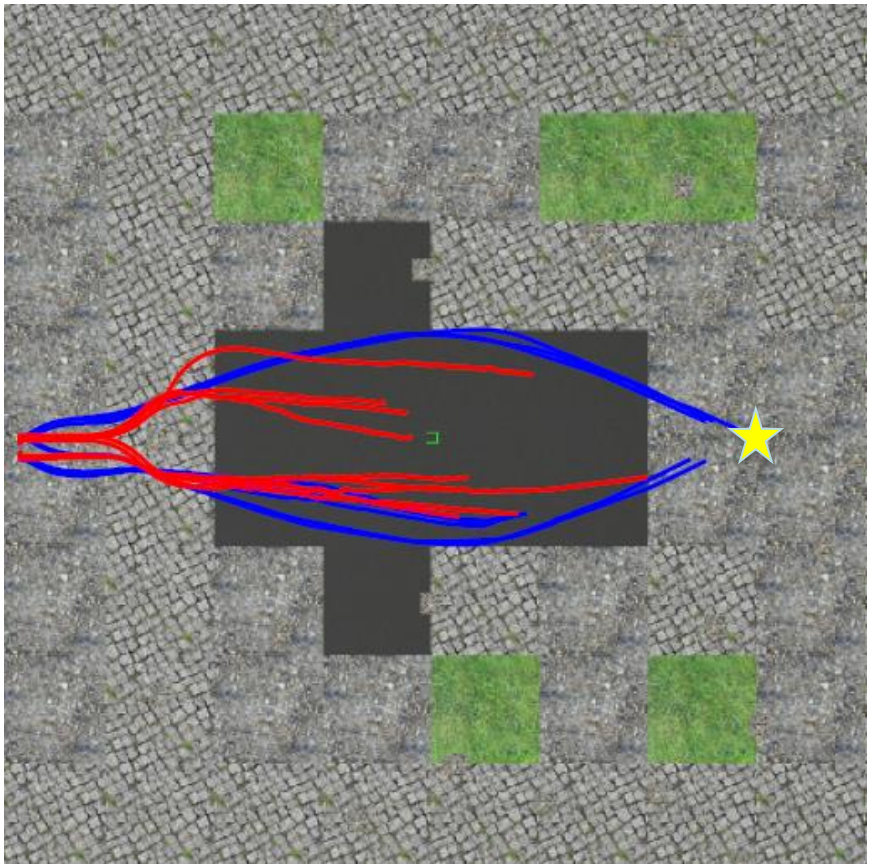
Using Fisher information from the localization process

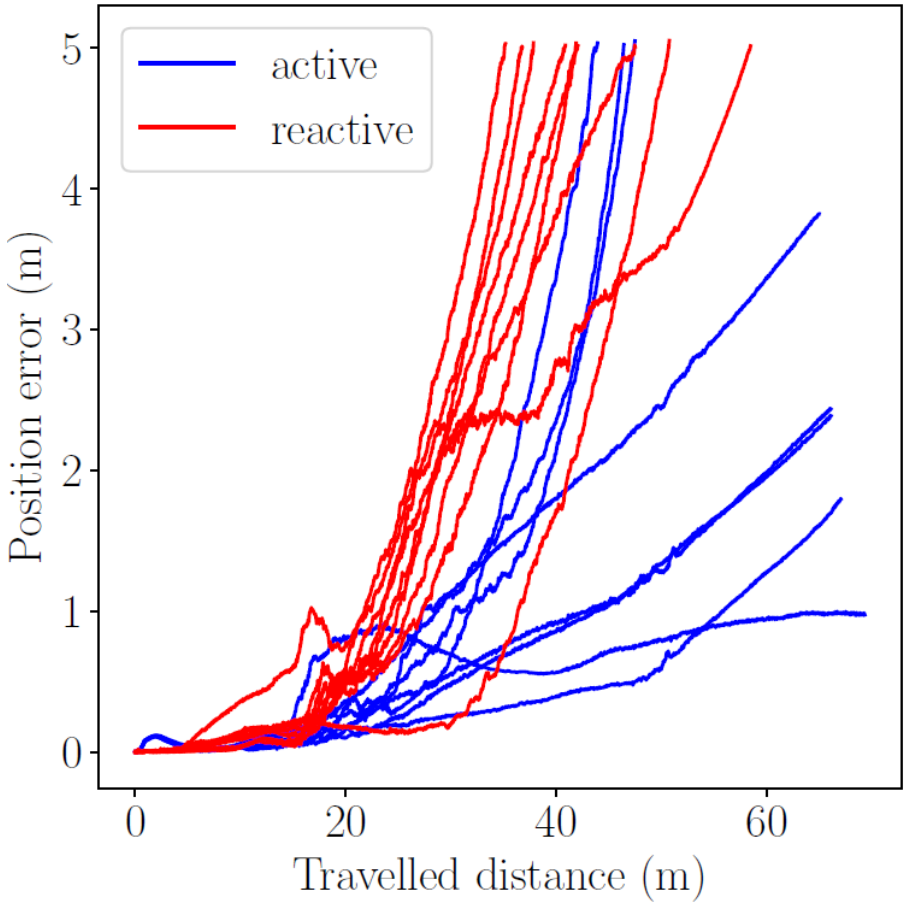
$$\text{Maximize } Reward = f(R_{\mathbf{collision}}, R_{\mathbf{goal}}, R_{\mathbf{perception}})$$

Simulation

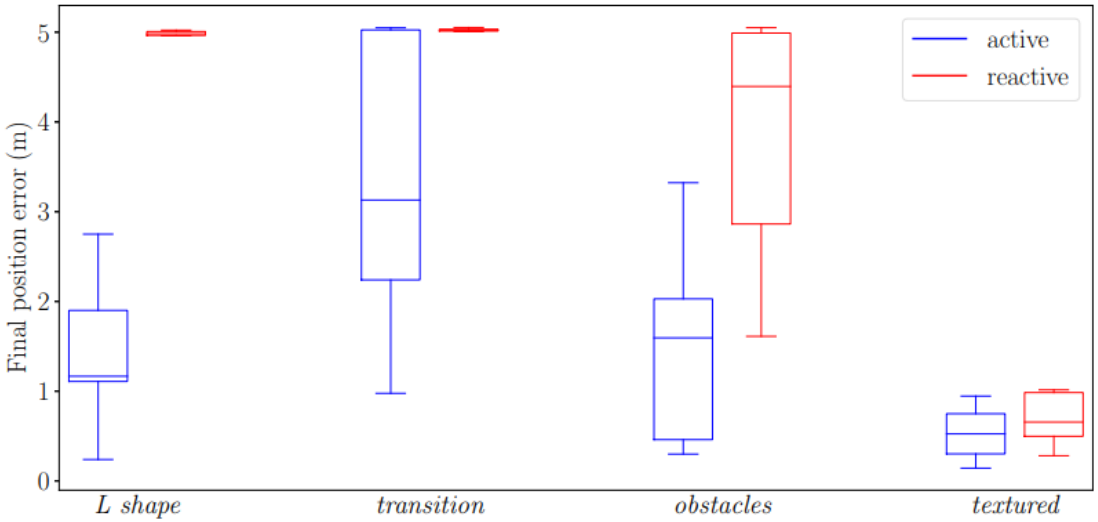
- Deal with textureless areas

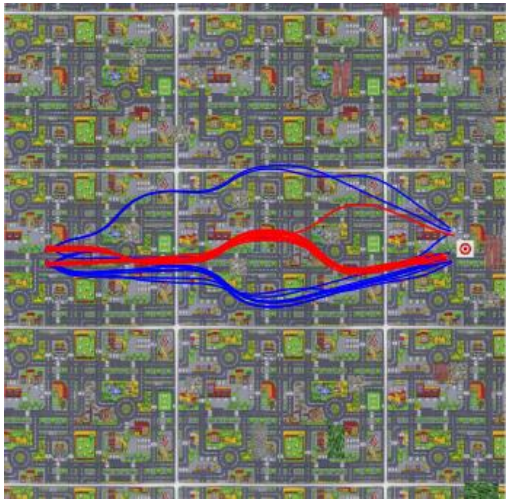
- **Perception-aware**
- **Purely-reactive**





Simulation: different environments





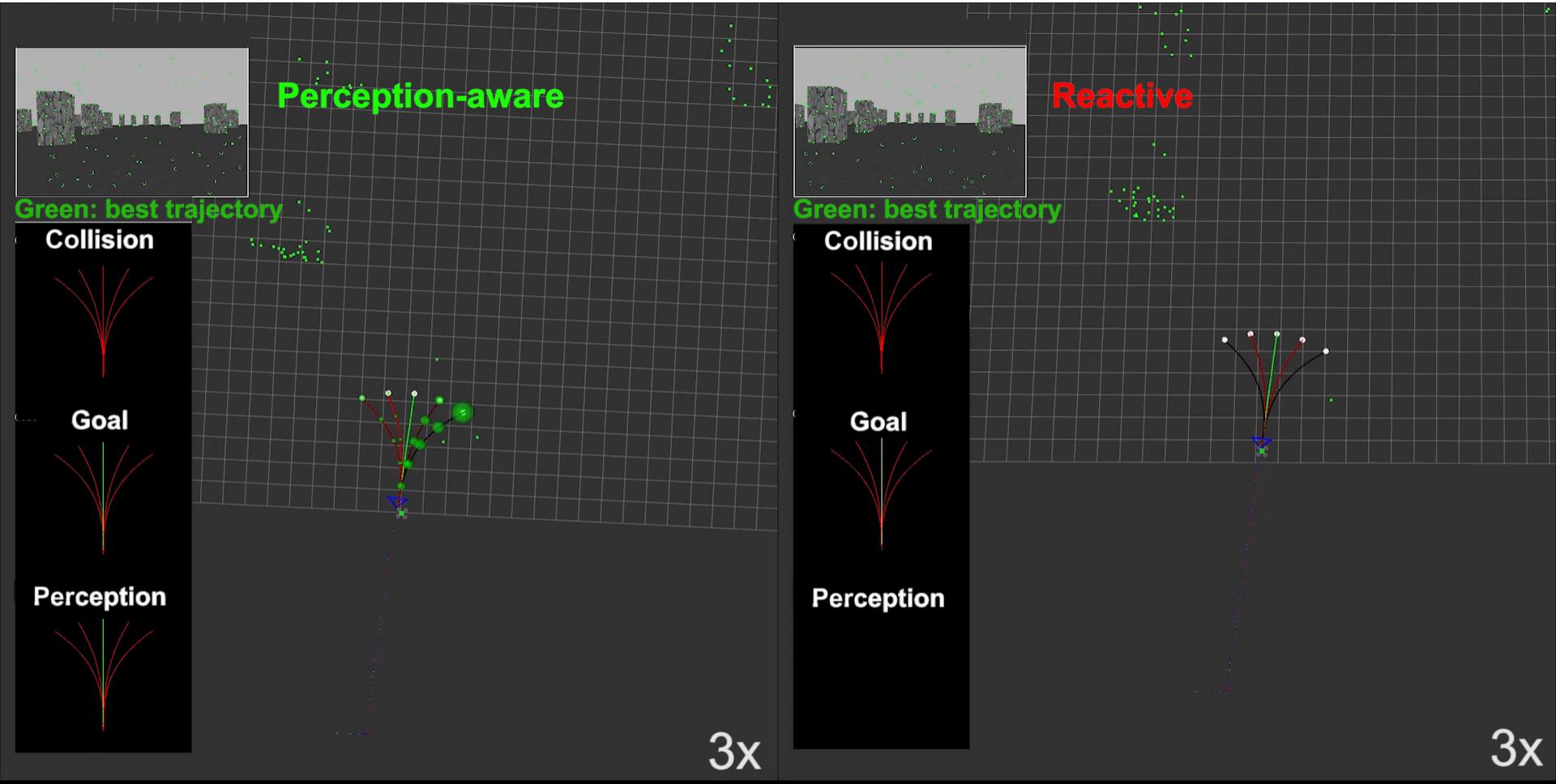
textured

Successes

Scene	Dimension (m)	Reactive	Ours
L shape	100 × 100	1	**5**
transition	100 × 100	0	**5**
obstacles	80 × 80	6	**9**
textured	60 × 60	9	**10**

- Able to deal with textureless regions
- Still beneficial in a scenario with no obvious “perception traps”

Obstacles: both attractor and repeller



In this sequence, our method commanded the quadrotor to fly close to obstacles because the features on the obstacles help to reduce state estimation uncertainty.

Collision and perception need to be considered together.

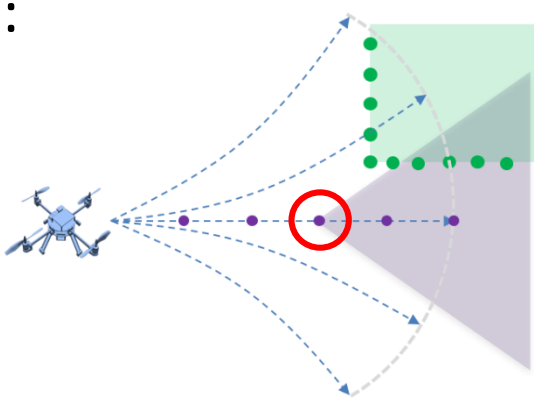
Balancing different costs in planning is not trivial (e.g., multi-objective optimization).

Map representation for active localization

Zhang et al, Beyond Point Clouds: Fisher Information Field for Active Visual Localization, *ICRA2019.*

- To quantify the localization quality for one pose:

1. Iterate over all the points in the FoV
2. Calculate the Fisher information from each point
3. Sum all the information and calculate some metrics

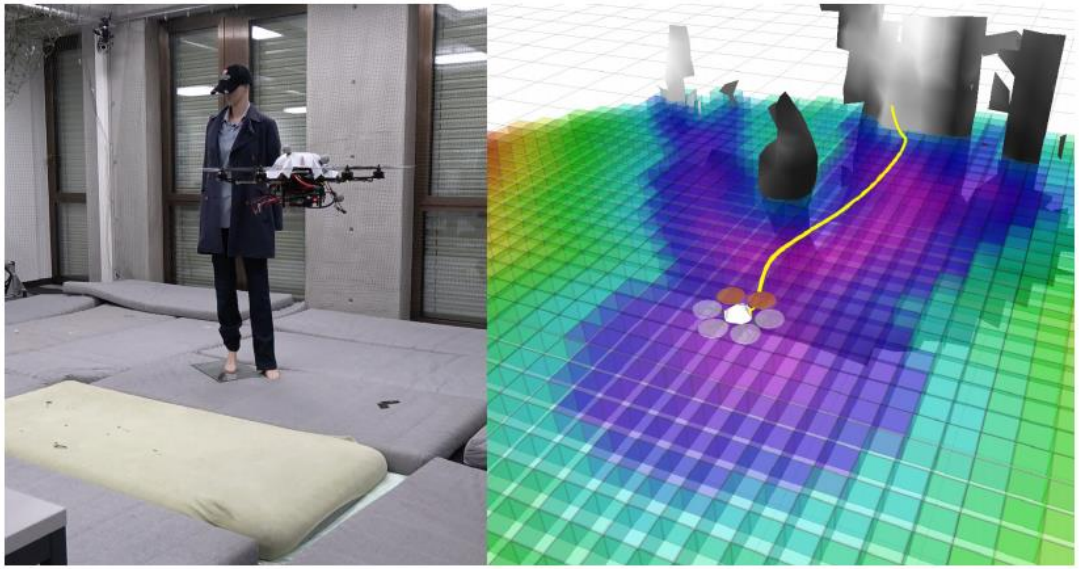


- Using point clouds from SLAM is convenient … but not that efficient
  - Linear with respect to the number of visible points
  - Cannot reuse the computed FIM at a pose
    - One planning session may need many iterations (e.g., trajectory optimization)
    - Multiple planning sessions in a known environment

    → **Need repeated evaluation of FIM in a known pose range.**

Motivation: dedicated map for planning

- Distance field for obstacle avoidance



Oleynikova et al, Voxblox: Incremental 3d euclidean signed distance fields for on-board mav planning, *IROS2017.*

Summarize the information of distance to obstacles in a compact format

- Efficiency
- Update online

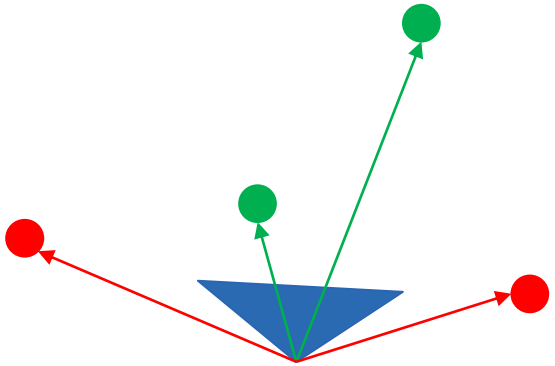
Can we have a dedicated and efficient map representation for the task of active visual localization/SLAM?

- **Goal**: given a pose, efficiently return its information (in terms of localization)
- How to encode the dependency on orientation?

Method: Understand the Visual Information

- The camera is a directional sensor with limited field of views



- **Idea**: Decompose the visual information for localization

Visual information for localization =

Geometric Relation + **Visibility Constraint**

Binary variable:
Visible or not

$$\mathtt{I}_{\mathtt{T}_{\mathtt{WC}}} = \sum_{k=1}^{N} v(\mathtt{T}_{\mathtt{WC}}, \mathbf{p}_i^{\mathtt{W}})\mathtt{I}_i$$

Camera is a directional sensor:
Does this depend on camera rotation?

Actual visibility is complicated:
Can we approximate it efficiently?

Method: Rotation Invariant FIM

$$\mathrm{I}_{\mathrm{T_{WC}}} = \sum_{k=1}^{N} v(\mathrm{T_{WC}}, \mathbf{p}_i^{\mathrm{W}}) \mathrm{I}_i$$

- FIM for i.i.d. Gaussian noise observation

$$\mathrm{I}_{\mathbf{x}}(\mathbf{z}) = \frac{1}{\sigma^2} (\mathrm{J}_{\mathbf{x}})^{\top} \mathrm{J}_{\mathbf{x}}, \quad \text{where } \mathrm{J}_{\mathbf{x}} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}}.$$

z: Pixel coordinates

x: Pose

- For camera observations
  1. Use bearing vectors

$$\mathrm{J}_i = \frac{\partial \mathbf{f}_i}{\partial \mathbf{p}_i^{\mathrm{C}}} \frac{\partial \mathbf{p}_i^{\mathrm{C}}}{\partial \mathrm{T_{WC}}}$$

f: bearing vector
p: 3d point
Twc: camera pose

  2. Use the derivative **in the world frame**

$$\frac{\partial \mathbf{p}_i^{\mathrm{C}}}{\partial \mathrm{T_{WC}}} \rightarrow \frac{\partial (\exp(\boldsymbol{\xi}^{\wedge}) \mathrm{T_{WC}})^{-1} \mathbf{p}_i^{\mathrm{W}}}{\partial \boldsymbol{\xi}} \text{ or } \frac{\partial (\mathrm{T_{WC}} \exp(\boldsymbol{\xi}^{\wedge}))^{-1} \mathbf{p}_i^{\mathrm{W}}}{\partial \boldsymbol{\xi}}$$

The actual FIM does not depend on the rotation

$$\mathrm{I}_i' = \frac{1}{\sigma^2} \mathrm{J}_i^{\top} \mathrm{R_{CC'}} \mathrm{R_{C'C}} \mathrm{J}_i = \mathrm{I}_i$$

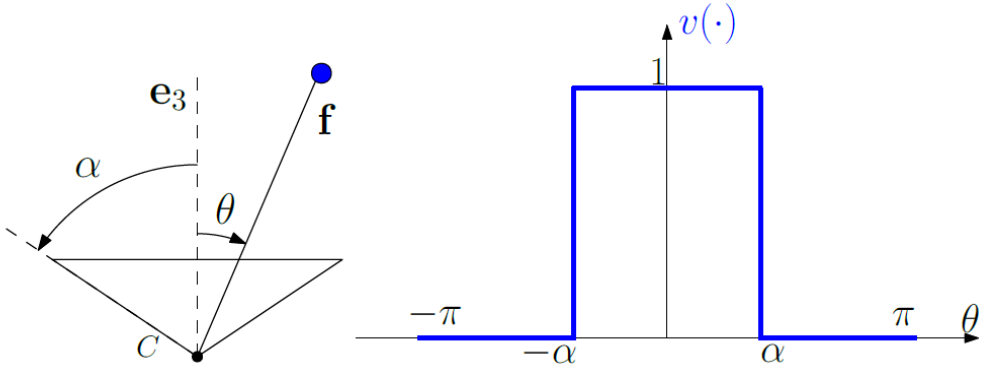
Intuition

- The observed bearing vectors only depend on the position → same information
- Parameterizing the invariant information in the a fixed frame (world frame).

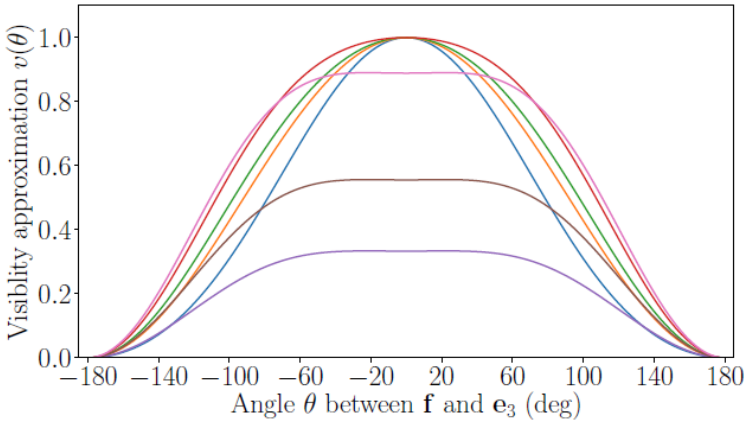
Method: Visibility Approximation

$$\mathtt{I}_{\mathtt{T}_{\mathtt{WC}}} = \sum_{k=1}^{N} v(\mathtt{T}_{\mathtt{WC}}, \mathbf{p}_i^{\mathtt{W}})\mathtt{I}_i$$

- Visibility as a quadratic function of $\cos\theta$



$$v(\theta_i) = a_2 \cos^2\theta_i + a_1 \cos\theta_i + a_0$$



Different coefficients

Advantages:

- Smoothness → suitable for optimization
- Can be expressed as dot product of the bearing vector and the optical axis

Problems:

- Limited expressive power
  - E.g., the long tail

Better (more expensive) approximation can also be used.

Method: Factor out the rotation!

$$\mathtt{I}_{\mathtt{T}_{\mathtt{WC}}} = \sum_{k=1}^{N} v(\mathtt{T}_{\mathtt{WC}}, \mathbf{p}_i^{\mathtt{W}}) \mathtt{I}_i$$

- Rotation-Invariant FIM + Quadratic Visibility Approximation

$$\mathtt{I}_{\mathtt{T}_{\mathtt{WC}}} = \sum_{k=1}^{N} v(\mathtt{T}_{\mathtt{WC}}, \mathbf{p}_i^{\mathtt{W}}) \mathtt{I}_i \approx \sum_{i=1}^{N} a_2 \cos\theta_i^2 \mathtt{I}_i + a_1 \cos\theta_i \mathtt{I}_i + a_0 \mathtt{I}_i$$

→ Rotation invariant

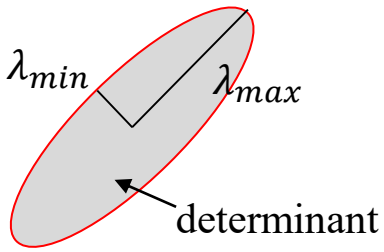
$$\mathtt{I}_{\mathtt{T}_{\mathtt{WC}}} \approx \sum_{i=1}^{N} \mathtt{A}(\mathtt{R}_{\mathtt{WC}}) \mathtt{H}(\mathbf{p}_i^{\mathtt{W}}, \mathbf{t}_{\mathtt{WC}}) \mathtt{B}(\mathtt{R}_{\mathtt{WC}}) \approx \mathtt{A}(\mathtt{R}_{\mathtt{WC}}) \left(\sum_{i=1}^{N} \mathtt{H}(\mathbf{p}_i^{\mathtt{W}}, \mathbf{t}_{\mathtt{WC}})\right) \mathtt{B}(\mathtt{R}_{\mathtt{WC}}).$$

The information kernel

- Does not depend on rotation → volumetric map
- Sum of the contributions of landmarks → incremental update
- Full FIM can be recovered by linear operation → efficient query

Method: All you need is a scalar metric

- Typical use case: FIM ➔ some scalar metric



Why not directly recover the metric?

- Matrix trace: T-optimality (related to $\|J\|_{\mathrm{F}}^2$)

$$\mathrm{Tr}(\mathrm{I}_{\mathrm{T}_{\mathrm{wc}}}) \approx \sum_{i=1}^{N} \mathrm{A}_{\mathrm{Tr}}(\mathrm{R}_{\mathrm{wc}})\mathrm{G}(\mathbf{p}_i^{\mathrm{w}}, \mathbf{t}_{\mathrm{wc}})\mathrm{B}_{\mathrm{Tr}}(\mathrm{R}_{\mathrm{wc}})$$

Trace kernel: $\mathrm{K}_{\mathrm{Tr}}(\mathbf{t}_{\mathrm{wc}}) = \sum_{i=1}^{N} \mathrm{G}(\mathbf{p}_i^{\mathrm{w}}, \mathbf{t}_{\mathrm{wc}})$

- More efficient, less memory
- Loss of information

Results: Efficiency

Setup

- 5m x 5m x 3m, 10cm voxel size → 75,000 voxels
- 5000 points, each pose have 0.5% of all points visible (25 points)
- Query the information/trace 500 K times

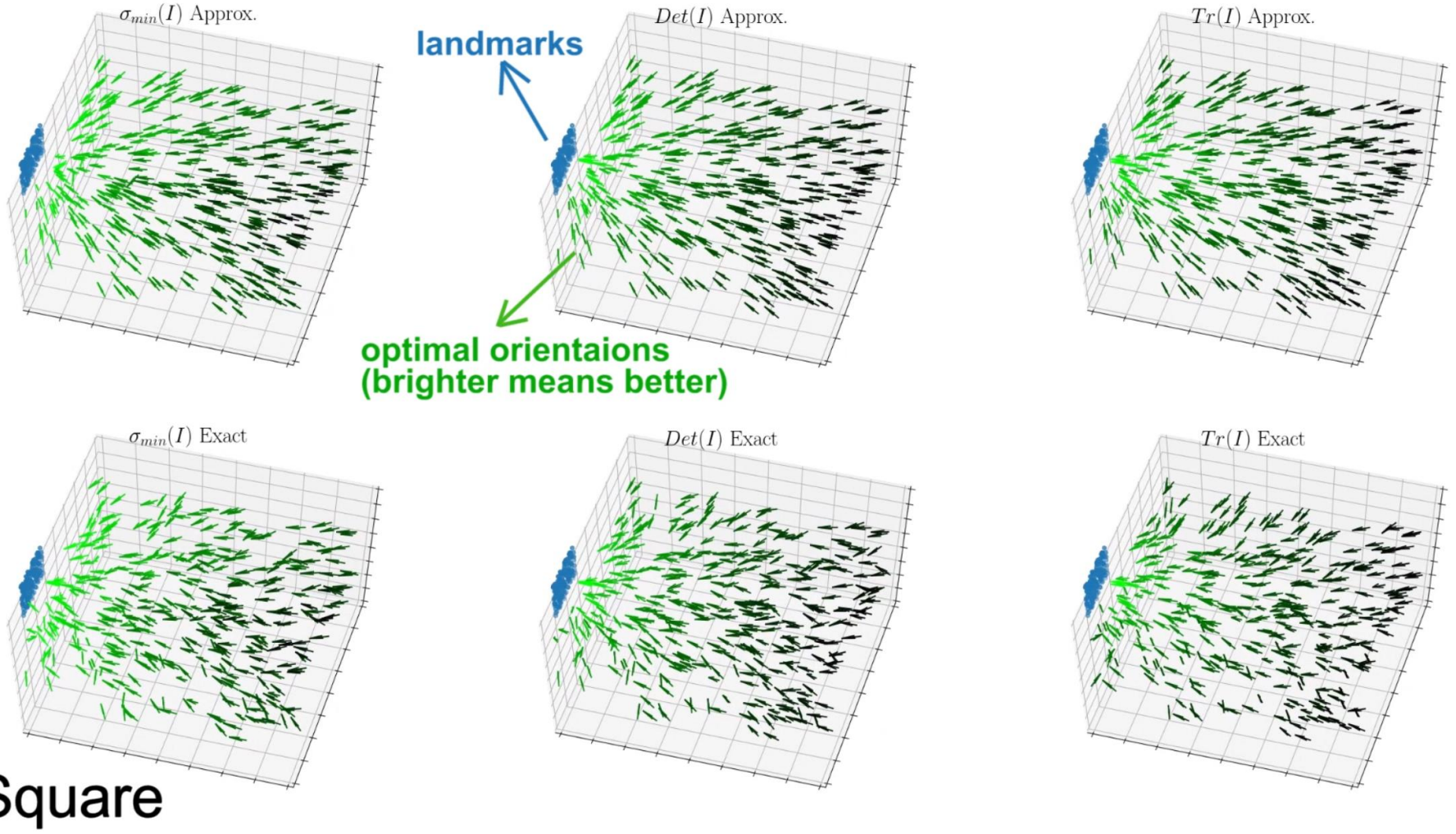
Method	$t_{\texttt{build}}$ (s)	$t_{\texttt{query}}$ (s)	Memory (MB)
Information Kernels	~ 1.4	~ 0.4	~ 270
Trace Kernels	~ 0.3	~ 0.05	~ 8
Point Cloud	0	~ 2	~ 0.11

Much shorter query time, with extra memory:

- Constant time vs. Linear with respect to the number of points
- Suitable to store as a persistent map

Results: Visualize the optimal views

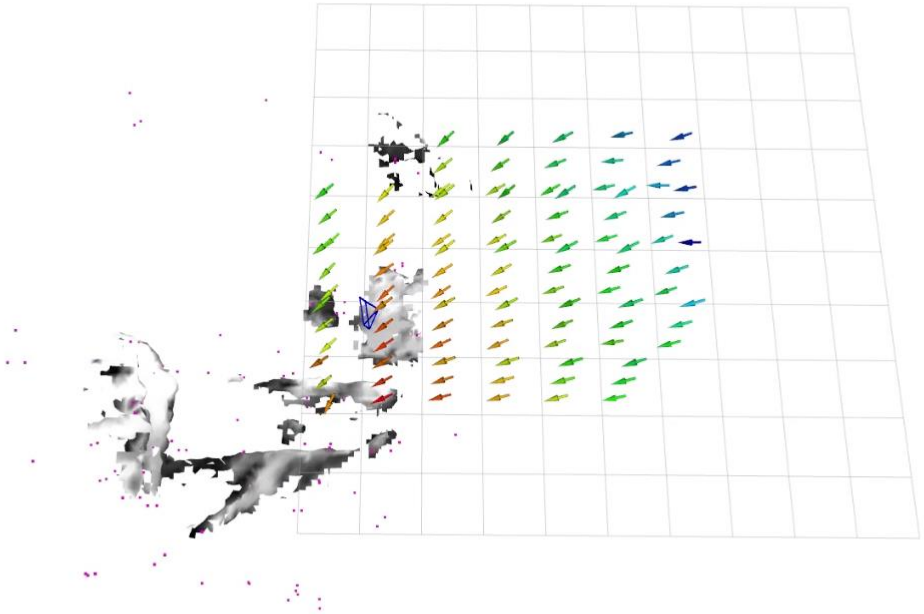
Top: information field; **Bottom**: exact information



Map representation for active localization



VIO image
green: landmarks



EuRoC MH05

localization quality

poor good

3X

Outline

- Vision-based state estimation
  - A short introduction to visual SLAM
  - Vision-based state estimation for mobile robots
- Perception-aware motion planning
  - Fisher information and its application
  - Receding horizon planning and map representation
- Outlook

(Just some opinions on future research directions)

Outlook

- SLAM + Learning
  - + semantic, high level information
    - (many papers nowadays)

Rosinol et al, Kimera: an Open-Source Library for Real-Time Metric-Semantic Localization and Mapping*, 2019.*

  - Provide (strong) priors for conventional SLAM modules

Tateno et al, CNN-SLAM: Real-time dense monocular slam with learned depth prediction, CVPR 2017

Yang et al, Deep virtual stereo odometry: leveraging deep depth prediction for monocular direct sparse odometry, ECCV 2018

Brachmann et al, Neural-guided RANSAC: Learning where to sample model hypotheses, ICCV 2019

  - Deep features
    - many work from computer vision community
    - good performance, but still rather slow nowadays

- Introduce theoretically-grounded methods in SLAM systems
  - More principled methods in system design (less heuristics)

Adaptive keyframe selection for multiple cameras

Juichung et al, Adaptive SLAM for Arbitrary Multi-Camera Systems, *accepted at ICRA 2020.*

- Motivation: the more cameras, the better
  - The optimization backend naturally generalizes to multiple cameras
  - But the frontend design heavily depends on the camera configurations

  → **Ideally, a SLAM pipeline should adapt to different camera configurations automatically (without the need to modify the code).**

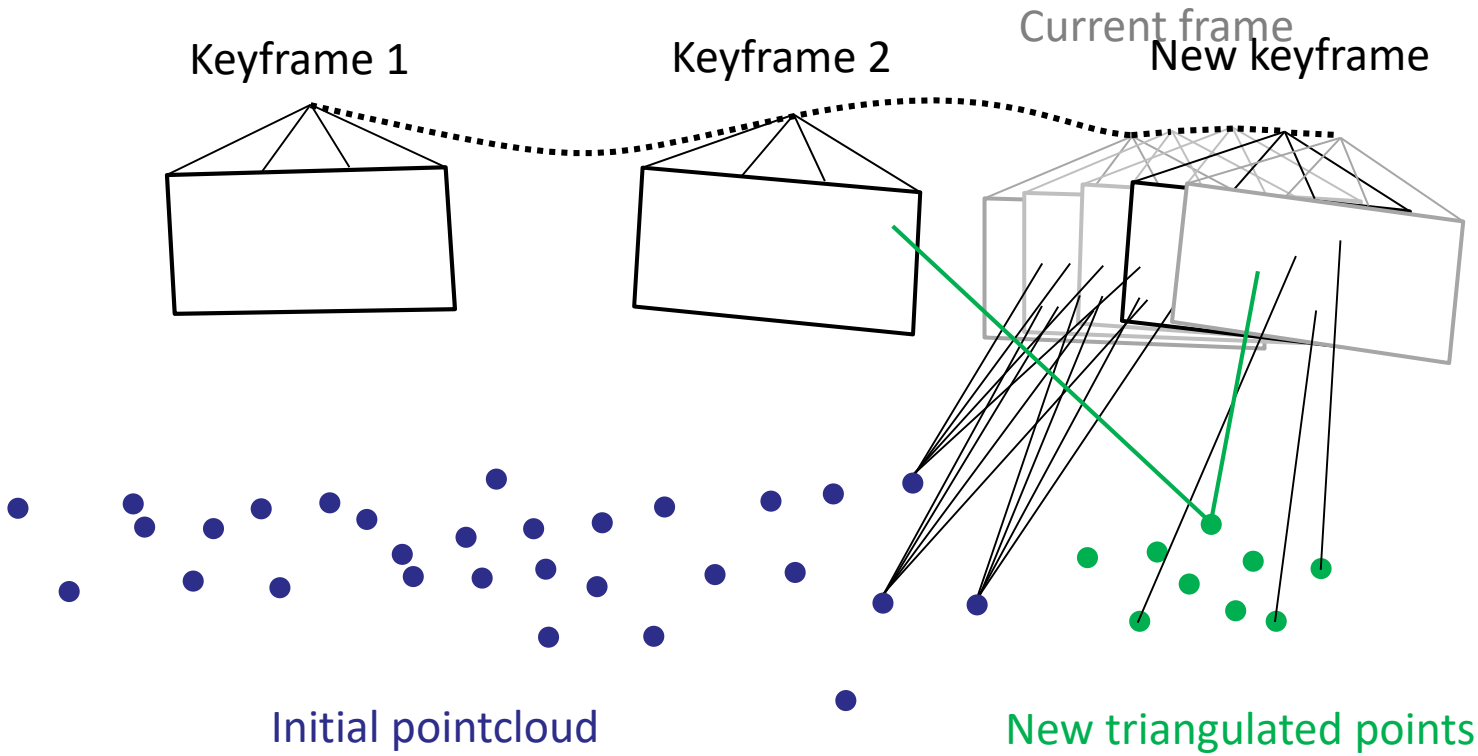
- Keyframe selection usually depends on heuristics (number of features, etc), which is not trivial to extend to multiple cameras:
  - From which camera to count the features?
  - Overlapping field-of-view?
  - …

Idea: use a metric that **does not** depends on the camera configuration as the criterion for keyframe selection.

Adaptive keyframe selection for multiple cameras

Juichung et al, Adaptive SLAM for Arbitrary Multi-Camera Systems, *accepted at ICRA 2020.*

- Back to the basics of visual odometry



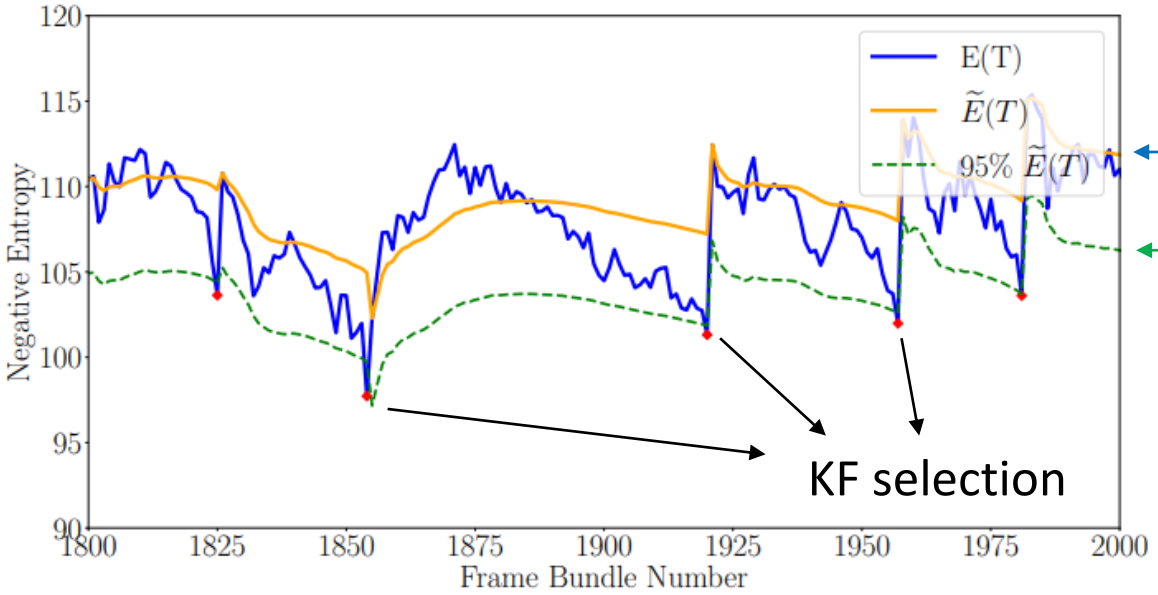
A keyframe is selected **when the map is not enough for localization**

→ When the localization uncertainty is high (the information for localization is low)

Adaptive keyframe selection for multiple cameras

Juichung et al, Adaptive SLAM for Arbitrary Multi-Camera Systems, *accepted at ICRA 2020.*

- **Method**: use the Fisher Information of the localization process (e.g., perspective-n-points) to determine whether to select a keyframe



Actual information (negative entropy here)

Threshold (estimated online) for keyframe selection

- **Advantages**:
  - **Just one parameter** for keyframe selection
  - Works for **arbitrary cameras** as long as the extrinsic are given

Outlook

- SLAM + Learning
  - + semantic, high level information
  - Provide (strong) priors for conventional SLAM modules
  - Deep features
- Introduce theoretically-grounded methods in SLAM systems
  - More principled methods in system design (less heuristics)
  - Algorithm guarantee and verification
    - (some work from Carlone's lab @ MIT)
- Map representation
  - Seems to be a central concept in both perception and planning
  - What would be an ideal map presentation for robotic applications?
    - Suitable for localization, planning; efficient; allow online update …
    - Should it be learnt or hand-crafted?

Bloesch et al, CodeSLAM: learning a compact, optimisable representation for dense visual SLAM, CVPR 2018

Ramos et al, Hilbert maps: scalable continuous occupancy mapping with stochastic gradient descent, IJRR 2016

Srivastava et al, Efficient, multifidelity perceptual representations via hierarchical gaussian mixture models , TRO 2018

……

Thanks! Questions?

References (available here)

VO Frontend

- Christian Forster, **Zichao Zhang**, Michael Gassner, Manuel Werlberger, Davide Scaramuzza. SVO: Semi-Direct Visual Odometry for Monocular and Multi-Camera Systems. *IEEE Transactions on Robotics,* Vol. 33, Issue 2, pages 249-265, Apr. 2017.
- Ruben Gomez-Ojeda, **Zichao Zhang**, Javier Gonzalez-Jimenez, Davide Scaramuzza, Learning-based Image Enhancement for Visual Odometry in Challenging HDR Environments, *IEEE International Conference on Robotics and Automation (ICRA),* 2018.
- **Zichao Zhang**, Christian Forster, Davide Scaramuzza, Active Exposure Control for Robust Visual Odometry in HDR Environments*, IEEE International Conference on Robotics and Automation (ICRA),* 2017.
- **Zichao Zhang**, Henri Rebecq, Christian Forster, Davide Scaramuzza, Benefit of Large Field-of-View Cameras for Visual Odometry, *IEEE International Conference on Robotics and Automation (ICRA)*, 2016. Dataset

Optimization Backend

- **Zichao Zhang**, Guillermo Gallego, Davide Scaramuzza. On the Comparison of Gauge Freedom Handling in Optimization-based Visual-Inertial State Estimation. *IEEE Robotics and Automation Letters (RA-L)*, 2018.

SLAM Evaluation

- **Zichao Zhang**, Davide Scaramuzza, A Tutorial on Quantitative Trajectory Evaluation for Visual(-inertial) Odometry, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018. Code
- **Zichao Zhang**, Davide Scaramuzza, Rethinking Trajectory Evaluation for SLAM: a Probabilistic, Continuous-Time Approach. *ICRA2019 Workshop on Dataset Generation and Benchmarking of SLAM Algorithms for Robotics and VR/AR*. **Best Paper Award.**

Active SLAM

- **Zichao Zhang**, Davide Scaramuzza, Beyond Point Clouds: Fisher Information Field for Active Visual Localization, *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.
- **Zichao Zhang**, Davide Scaramuzza, Perception-aware Receding Horizon Navigation for MAVs, *IEEE International Conference on Robotics and Automation (ICRA),* 2018.